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Next item →

1. Which variable(s) are propagated as square roots in the square-root Kalman filter (versus their non-square-root equivalents in the standard Kalman filter)? Select all that apply.

1 / 1 point

- ☐ The predicted state $\hat{x}_k^-$.
- ☒ The prediction-error covariance matrix $\Sigma_{x,k}^-$.

✔ Correct
Yes. This is propagated as a square-root quantity.

- ☐ The estimated state $\hat{x}_k^+$.
- ☒ The estimation-error covariance matrix $\Sigma_{x,k}^+$.

✔ Correct
Yes. This is propagated as a square-root quantity.

2. What is the primary motivation behind the development of the square-root Kalman filter?

1 / 1 point

- ☐ It is faster than the standard Kalman filter (i.e., it requires fewer floating-point operations per iteration).
- ☒ It reduces the number of bits per variable required to estimate the state to a desired precision.
- ☐ It forces all covariance matrices to be positive definite.
- ☐ It provides more accurate estimates than the standard Kalman filter.

✔ Correct
Yes, this is true. Alternately, it increases the precision for a given number of bits of storage per variable.

3. Run the Kalman-filter and square-root Kalman-filter code in the Coursera Labs Jupyter notebook for this lesson. What do you observe?

1 / 1 point

- ☒ There is no noticeable difference in results between the two filters.
- ☐ The standard Kalman filter is noticeably less accurate than the square-root Kalman filter.
- ☐ The square-root Kalman filter is noticeably less accurate than the standard Kalman filter.
- ☐ There are noticeable differences near the beginning of the simulation, but after some time the square-root Kalman filter converges to the standard Kalman-filter solution.

✔ Correct
Yes. When using double-precision floating-point variables in Octave, there is no noticeable difference in results.